

Updated Research Question 1

Almost-Balanced Positive Dichotomous Allocations and the Negative Dichotomous Subsidy Problem

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1 The negative dichotomous problem

Let $N = [n]$ be the set of agents and let M be a set of m indivisible chores. Agent i has a cost function

$$c_i : 2^M \rightarrow \mathbb{Z}_{\geq 0}, \quad c_i(\emptyset) = 0,$$

with dichotomous marginals:

$$c_i(S \cup \{g\}) - c_i(S) \in \{0, 1\} \quad \text{for every } S \subseteq M \text{ and } g \notin S.$$

Equivalently, the corresponding valuation $-c_i$ has marginal values in $\{0, -1\}$.

For a complete allocation $\mathcal{A} = (A_1, \dots, A_n)$ and a nonnegative subsidy vector $p \in \mathbb{R}_{\geq 0}^n$, envy-freeness requires

$$c_i(A_i) - p_i \leq c_i(A_j) - p_j \quad \text{for every } i, j \in N.$$

The target conjecture is the negative analogue of the positive dichotomous theorem of Barman–Krishna–Narahari–Sadhukhan (BKNS).

Conjecture 1 (Negative dichotomous 0/1-subsidy conjecture). *For every negative dichotomous instance, there exists a complete allocation $\mathcal{A}$ and a subsidy vector*

$$p \in \{0, 1\}^n$$

such that

$$c_i(A_i) - p_i \leq c_i(A_j) - p_j \quad \forall i, j \in N.$$

Ideally, such a solution should also be computable in polynomial time in the value-oracle model.

The purpose of this note is to formulate a strengthened positive-dichotomous statement which, if proved, would imply Conjecture 1. The strengthening is weaker than exact balancedness: bundles need only differ in size by at most one, together with a compatibility condition between bundle size and subsidy.

2 The size-shift transformation

Given the negative dichotomous cost functions c_i , define a positive valuation

$$\tilde{v}_i(S) := |S| - c_i(S), \quad S \subseteq M.$$

For every $g \notin S$,

$$\begin{aligned} \tilde{v}_i(S \cup \{g\}) - \tilde{v}_i(S) &= 1 - (c_i(S \cup \{g\}) - c_i(S)) \\ &\in \{0, 1\}. \end{aligned}$$

Hence each $\tilde{v}_i$ is a positive dichotomous valuation of exactly the type considered in the BKNS paper.

For an allocation $\mathcal{B} = (B_1, \dots, B_n)$, let

$$\tilde{w}_{\mathcal{B}}(i, j) := \tilde{v}_i(B_j) - \tilde{v}_i(B_i)$$

denote the envy-graph arc weight in the transformed positive instance, and let

$$w_{\mathcal{B}}^c(i, j) := c_i(B_i) - c_i(B_j)$$

denote the corresponding envy-graph arc weight in the chore instance. A direct calculation gives

$$w_{\mathcal{B}}^c(i, j) = \tilde{w}_{\mathcal{B}}(i, j) + |B_i| - |B_j|. \quad (1)$$

Thus the only difference between the two envy graphs is the potential term $|B_i| - |B_j|$.

If $(\mathcal{B}, q)$ is envy-free in the transformed positive instance, then

$$\tilde{w}_{\mathcal{B}}(i, j) \leq q_i - q_j \quad \forall i, j. \quad (2)$$

Combining (1) and (2) gives

$$w_{\mathcal{B}}^c(i, j) \leq (q_i + |B_i|) - (q_j + |B_j|). \quad (3)$$

This inequality identifies exactly what must be controlled in addition to the BKNS 0/1 subsidy bound.

3 Almost-balanced allocations

Write

$$m = kn + r, \quad 0 \leq r < n.$$

An almost-balanced allocation has bundle sizes

$$|B_i| \in \{k, k+1\} \quad \forall i.$$

Necessarily, exactly r agents receive $k+1$ goods and the remaining $n-r$ agents receive k goods.

Define

$$L := \{i \in N : |B_i| = k\}, \quad H := \{i \in N : |B_i| = k+1\}.$$

We refer to L as the set of *small-bundle agents* and H as the set of *large-bundle agents*.

The proposed strengthening of BKNS is the following.

Research Question 1 (Almost-balanced positive dichotomous subsidy problem). *For every positive dichotomous instance with m goods and n agents, does there exist a complete allocation $\mathcal{B} = (B_1, \dots, B_n)$ and a subsidy vector $q \in \{0, 1\}^n$ such that:*

- (i) $(\mathcal{B}, q)$ is envy-free for the positive instance;
 - (ii) $||B_i| - |B_j|| \leq 1$ for all i, j ;
 - (iii) whenever $|B_i| > |B_j|$, one has
- $$q_i \leq q_j. \quad (4)$$

Can such an allocation and subsidy vector be found in polynomial time?

Condition (4) says that an agent receiving a larger bundle should never receive a strictly larger subsidy than an agent receiving a smaller bundle. In the notation above,

$$q_h \leq q_\ell \quad \forall h \in H, \ell \in L.$$

Since the subsidies are binary, this is a very simple two-tier compatibility condition.

4 Why an affirmative answer solves the negative problem

Proposition 1. *An affirmative answer to Research Question 1 implies Conjecture 1. The implication is constructive: a polynomial-time algorithm for Research Question 1 yields a polynomial-time algorithm for the negative dichotomous problem.*

Proof. Start from an arbitrary negative dichotomous instance $\{c_i\}_{i \in N}$ on the original set M of m chores. Define the transformed positive dichotomous instance by

$$\tilde{v}_i(S) = |S| - c_i(S).$$

Assume Research Question 1 is true. Then there exists a complete almost-balanced allocation $\mathcal{B} = (B_1, \dots, B_n)$ and $q \in \{0, 1\}^n$ such that $(\mathcal{B}, q)$ is envy-free for the transformed positive instance and

$$q_h \leq q_\ell \quad \forall h \in H, \ell \in L.$$

Define

$$d_i := |B_i| - k \in \{0, 1\}.$$

Thus $d_i = 0$ for $i \in L$ and $d_i = 1$ for $i \in H$. Consider the numbers

$$r_i := q_i + d_i.$$

The four possible combinations are

$ B_i $	q_i	$r_i = q_i + d_i$
k	0	0
k	1	1
$k+1$	0	1
$k+1$	1	2

The only way that the set $\{r_i : i \in N\}$ can contain both 0 and 2 is if there exist $\ell \in L$ and $h \in H$ with

$$q_\ell = 0, \quad q_h = 1.$$

This is ruled out by the compatibility condition $q_h \leq q_\ell$. Therefore

$$\max_i r_i - \min_i r_i \leq 1.$$

Let

$$\alpha := \min_i r_i$$

and define

$$p_i := r_i - \alpha = q_i + d_i - \alpha.$$

Then $p_i \in \{0, 1\}$ for every agent.

Moreover, subtraction of the common constant α does not affect differences, so

$$p_i - p_j = (q_i - q_j) + (d_i - d_j).$$

Since $d_i - d_j = |B_i| - |B_j|$, inequality (3) yields

$$\begin{aligned} w_{\mathcal{B}}^c(i, j) &\leq (q_i - q_j) + |B_i| - |B_j| \\ &= p_i - p_j. \end{aligned}$$

Equivalently,

$$c_i(B_i) - p_i \leq c_i(B_j) - p_j \quad \forall i, j.$$

Thus $(\mathcal{B}, p)$ is an envy-free solution of the original negative dichotomous instance with

$$p \in \{0, 1\}^n.$$

This proves Conjecture 1.

The transformation requires only value queries and elementary arithmetic. Hence if the almost-balanced positive allocation and its subsidy vector can be found in polynomial time, so can the resulting negative-dichotomous solution. $\square$

Remark 1 (No dummy chores are needed). *This is an advantage over the exact-balanced formulation. Exact balance would require m to be divisible by n , or else the introduction of dummy zero-cost chores. Almost-balancedness is available for every pair (m, n) , with the canonical bundle sizes k and $k + 1$.*

5 The exact weaker condition in terms of paths

The subsidy compatibility condition in Research Question 1 is convenient, but it is stronger than necessary. The size-shift identity gives an exact graph-theoretic condition.

For an envy-freeable positive allocation $\mathcal{B}$, let

$$\tilde{d}_{\mathcal{B}}(i, j)$$

denote the maximum weight of a directed path from i to j in the positive envy graph. Along a path from i to j , the cardinality terms telescope, giving

$$d_{\mathcal{B}}^c(i, j) = \tilde{d}_{\mathcal{B}}(i, j) + |B_i| - |B_j|. \quad (5)$$

Assume the transformed positive allocation already has minimum per-agent subsidy at most 1. Equivalently, every positive-graph path starting from any agent has weight at most 1.

For an almost-balanced allocation, there are only three possibilities for the cardinality correction:

$$|B_i| - |B_j| \in \{-1, 0, 1\}.$$

If $i \in L$ and $j \in H$, the correction is -1 , so there is no danger of exceeding 1. If i, j belong to the same size tier, the correction is 0. The only potentially problematic case is

$$i = h \in H, \quad j = \ell \in L,$$

for which the correction equals $+1$.

Therefore, conditional on the positive 0/1 subsidy property, the chore allocation has subsidy at most 1 if and only if

$$\tilde{d}_{\mathcal{B}}(h, \ell) \leq 0 \quad \forall h \in H, \ell \in L. \quad (6)$$

In words:

There must be no positive-weight directed path in the transformed positive envy graph from a large-bundle agent to a small-bundle agent.

The subsidy compatibility condition is an easy sufficient condition for (6). Indeed, if $(\mathcal{B}, q)$ is envy-free, then every path from h to ℓ has weight at most $q_h - q_\ell$. Thus

$$q_h \leq q_\ell \implies \tilde{d}_{\mathcal{B}}(h, \ell) \leq 0.$$

Hence there are two useful formulations of the strengthened positive problem:

- (a) the *exact path version*: require (6);
- (b) the *simpler subsidy version*: require $q_h \leq q_\ell$ for all $h \in H, \ell \in L$.

The second is stronger but may be considerably easier to maintain algorithmically.

6 A stronger condition tailored to the IISc algorithm

Let

$$M(q) := \{i \in N : q_i \geq q_j \text{ for every } j \in N\}$$

denote the maximally subsidised agents, exactly as in the BKNS paper. Since $q \in \{0, 1\}^n$, a particularly convenient sufficient condition for the compatibility requirement is

$$L \subseteq M(q). \tag{7}$$

That is, every agent with a minimum-cardinality bundle is maximally subsidised.

To see why this suffices, there are two cases. If $\max_i q_i = 0$, then all subsidies are zero and the compatibility condition is automatic. If $\max_i q_i = 1$, then (7) gives

$$q_\ell = 1 \quad \forall \ell \in L,$$

while of course $q_h \leq 1$ for every $h \in H$. Therefore

$$q_h \leq q_\ell \quad \forall h \in H, \ell \in L.$$

This suggests an algorithm-friendly strengthening of Research Question 1.

Algorithmic Target (IISc-style invariant version). *Can the BKNS incremental algorithm be modified so that, throughout the construction, it maintains all three invariants*

- (I1) $\mathcal{B}$ is envy-freeable and its minimum subsidy vector $q \in \{0, 1\}^n$;
- (I2) $\max_i |B_i| - \min_i |B_i| \leq 1$;
- (I3) $\{i : |B_i| = \min_j |B_j|\} \subseteq M(q)$?

If these invariants can be maintained until every good is allocated, then the final allocation satisfies Research Question 1, and therefore solves the negative dichotomous problem by Proposition 1.

7 Why this may fit the original BKNS proof architecture

The original positive-dichotomous algorithm proceeds incrementally. At each step it allocates one previously unallocated good and, when necessary, permutes the existing bundles. Its central invariant is that the current partial allocation is envy-freeable with subsidy vector in $\{0, 1\}^n$. A distinguished role is played by the maximally subsidised set $M(q)$.

The almost-balanced formulation suggests adding only a cardinality discipline to that framework. At any stage the bundle sizes should be either k or $k + 1$ for the current value of k . While distributing the next block of at most n goods,

- bundles already given their next good have size $k + 1$;
- bundles still waiting have size k ;
- the desired new invariant says that the agents still waiting for their next good should remain maximally subsidised.

This is strikingly close to the logic already present in the BKNS EXTEND step, where an important case is handled by assigning the new good after identifying a suitable maximally subsidised agent (possibly after a permutation of bundles). The main technical question is therefore not whether the old algorithm can preserve 0/1 subsidies—that is already known—but whether its choices and bundle permutations can be restricted so as to preserve both almost-balancedness and condition (I3).

The proposed proof program is consequently:

- Step 1:** Revisit the BKNS extendable case under the additional restriction that a new good should be assigned only to a minimum-cardinality bundle whenever the allocation is not currently balanced.
- Step 2:** Determine whether the required bundle permutation can always be chosen so that the receiver remains in $M(q)$ before insertion and the minimum-cardinality agents remain in $M(q)$ afterwards.
- Step 3:** Reanalyse the non-extendable/FINDSINK case with the same cardinality restriction. This is likely to be the genuinely difficult part.
- Step 4:** If the strong invariant (I3) fails, weaken it to the exact path condition (6): no positive-weight path from a large-bundle agent to a small-bundle agent.

8 Summary of the updated research question

The updated Research Question 1 can be stated compactly as follows.

Updated Research Question 1. For every positive dichotomous instance, can one find an envy-free solution $(\mathcal{B}, q)$ such that $q \in \{0, 1\}^n$, the allocation is almost balanced,

$$||B_i| - |B_j|| \leq 1 \quad \forall i, j,$$

and a larger-bundle agent never receives a strictly larger subsidy than a smaller-bundle agent,

$$|B_i| > |B_j| \implies q_i \leq q_j?$$

A stronger but algorithmically natural sufficient condition is that every minimum-cardinality agent belongs to $M(q)$. An exact weaker formulation is that the transformed positive envy graph contain no positive-weight path from a large-bundle agent to a small-bundle agent.

Any affirmative answer implies the 0/1-subsidy conjecture for negative dichotomous valuations through the size-shift transformation. The almost-balanced formulation has the further advantage that it works directly for arbitrary m and n , without adding dummy chores, and it appears substantially closer to the incremental architecture of the original IISc paper than the exact-balanced formulation.

Context. This note develops the size-shift idea in the working note on negative dichotomous valuations and the 0/1-subsidy machinery from Barman, Krishna, Narahari and Sadhukhan, *Achieving Envy-Freeness with Limited Subsidies under Dichotomous Valuations*.